1. Classical Sets and Fuzzy Sets

Let the universal set be

$$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

A fuzzy set A is given by

$$A = \{0.1/1,\ 0.3/2,\ 0.5/3,\ 0.7/4,\ 0.9/5,\ 1/6,\ 0.8/7,\ 0.6/8,\ 0.4/9,\ 0.2/10\}.$$

Find:

1. The **support** of A.
2. The **core** of A.
3. The **height** of A.
4. The α-**cut** of A for $\alpha = 0.5$.

2. Cartesian Product and Fuzzy Relations

Let

$$A = \{0.2/x_1,\ 0.6/x_2,\ 0.9/x_3\}$$

and

$$B = \{0.4/y_1,\ 0.7/y_2\}.$$

Find the fuzzy Cartesian product $A \times B$, where

$$\mu_{A\times B}(x, y) = \min[\mu_A(x), \mu_B(y)].$$

Represent your answer as a **membership matrix**.

3. Composition of Fuzzy Relations

Consider two fuzzy relations:

$$R = \begin{bmatrix} 0.2 & 0.5 & 0.8 \\ 0.7 & 0.4 & 0.3 \end{bmatrix}$$

and

$$S = \begin{bmatrix} 0.6 & 0.9 \\ 0.5 & 0.2 \\ 0.8 & 0.4 \end{bmatrix}.$$

Calculate the **max-min composition**

$$T = R \circ S.$$

4. Membership Function and Fuzzification

The fuzzy set "**Young Age**" is represented by the triangular membership function

$$\mu_Y(x) = \begin{cases} 1, & x \leq 20 \\ \dfrac{40 - x}{20}, & 20 < x < 40 \\ 0, & x \geq 40 \end{cases}$$

Find the membership values for ages:

$$18,\ 20,\ 25,\ 30,\ 35,\ 40,\ 45.$$

Hence, construct the corresponding fuzzy set.

5. Lambda-Cuts and Defuzzification

A fuzzy set representing temperature is

$$T = \{0.1/10,\ 0.3/20,\ 0.6/30,\ 0.8/40,\ 1.0/50,\ 0.7/60,\ 0.4/70\}.$$

Find:

1. The **0.4-cut**.
2. The **0.7-cut**.
3. The **0.9-cut**.
4. The **strong 0.7-cut**.

6. Fuzzy Arithmetic

Two fuzzy numbers are given by

$$A = \{0.2/1,\ 0.5/2,\ 0.8/3,\ 1/4,\ 0.6/5\}$$

and

$$B = \{0.3/2,\ 0.7/3,\ 1/4,\ 0.8/5\}.$$

Using the **extension principle**, calculate the fuzzy addition

$$C = A + B.$$

Give the resulting membership values for all possible sums.

7. Measures of Fuzziness and Fuzzy Integral

Consider the fuzzy set

$$A = \{0.2/x_1,\ 0.4/x_2,\ 0.7/x_3,\ 1.0/x_4,\ 0.6/x_5\}.$$

Calculate the **fuzziness measure** using

$$F(A) = \frac{1}{n}\sum_{i=1}^{n} 2\min(\mu_A(x_i), 1 - \mu_A(x_i)).$$

Also determine which element contributes the **maximum amount of fuzziness**.

Solutions:

Q1. Given: Universal set $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

Fuzzy set $A = \{0.1/1, 0.3/2, 0.5/3, 0.7/4, 0.9/5, 1/6, 0.8/7, 0.6/8, 0.4/9, 0.2/10\}$

1) Support of A

Formula: Support (A) = $\{x \in X \mid \mu_A(x) > 0\}$

From A, all elements have membership > 0.

$\therefore$ Support (A) = $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$

2) Core of A

Formula: Core (A) = $\{x \in X \mid \mu_A(x) = 1\}$

From A, $\mu_A(6) = 1$ and no other element has membership 1.

$\therefore$ Core (A) = $\{6\}$

3) Height of A

Formula: Height (A) = sup $\{\mu_A(x) \mid x \in X\}$

= maximum membership value in A

From A, the maximum membership value is 1 (at $x = 6$).

$\therefore$ Height (A) = 1

4) α-cut of A for $\alpha = 0.5$

Formula: $A_\alpha = \{x \in X \mid \mu_A(x) \geq \alpha\}$

Here $\alpha = 0.5 \Rightarrow A_{0.5} = \{x \in X \mid \mu_A(x) \geq 0.5\}$

From A, the elements with membership ≥ 0.5 are:

3 (0.5), 4 (0.7), 5 (0.9), 6 (1), 7 (0.8), 8 (0.6)

$\therefore$ $A_{0.5} = \{3, 4, 5, 6, 7, 8\}$

Summary of Answers:

1) Support (A) = $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
2) Core (A) = $\{6\}$
3) Height (A) = 1
4) $A_{0.5} = \{3, 4, 5, 6, 7, 8\}$

Q2. Cartesian Product and Fuzzy Relations

Given: $A = \{ 0.2/x_1 , 0.6/x_2 , 0.9/x_3 \}$

$B = \{ 0.4/y_1 , 0.7/y_2 \}$

To find: Fuzzy Cartesian Product $A \times B$

Formula: $\mu_{A \times B}(x_i , y_j) = \min [\mu_A(x_i) , \mu_B(y_j)]$

(Take minimum of membership values)

Step 1: List the membership values.

For set A

$\mu_A(x_1) = 0.2$

$\mu_A(x_2) = 0.6$

$\mu_A(x_3) = 0.9$

For set B

$\mu_B(y_1) = 0.4$

$\mu_B(y_2) = 0.7$

Step 2: Use the formula $\mu_{A \times B}(x_i , y_j) = \min[\mu_A(x_i), \mu_B(y_j)]$

Compute for each pair (x_i , y_j).

(x_1 , y_1) : $\min (0.2 , 0.4) = 0.2$

(x_1 , y_2) : $\min (0.2 , 0.7) = 0.2$

(x_2 , y_1) : $\min (0.6 , 0.4) = 0.4$

(x_2 , y_2) : $\min (0.6 , 0.7) = 0.6$

(x_3 , y_1) : $\min (0.9 , 0.4) = 0.4$

(x_3 , y_2) : $\min (0.9 , 0.7) = 0.7$

Step 3: Write the results in a membership matrix.

Rows → elements of A (x_1 , x_2 , x_3)

Columns → elements of B (y_1 , y_2)

Membership Matrix of $A \times B$

$A \times B$	y_1 (0.4)	y_2 (0.7)
x_1 (0.2)	0.2	0.2
x_2 (0.6)	0.4	0.6
x_3 (0.9)	0.4	0.7

Step 4: List the fuzzy Cartesian product as a set of ordered pairs (optional).

$A \times B = \{ 0.2/(x_1, y_1) , 0.2/(x_1, y_2) , 0.4/(x_2, y_1) , 0.6/(x_2, y_2) , 0.4/(x_3, y_1) , 0.7/(x_3, y_2) \}$

Final Answer: The membership matrix of $A \times B$ is

$A \times B$	y_1 (0.4)	y_2 (0.7)
x_1 (0.2)	0.2	0.2
x_2 (0.6)	0.4	0.6
x_3 (0.9)	0.4	0.7

* The max-min composition $T = R \circ S$ results in:

$$T = \begin{bmatrix} 0.8 & 0.4 \\ 0.6 & 0.7 \end{bmatrix}$$

Core Concept: What is Max-Min Composition?

In standard matrix multiplication, you multiply elements pair-by-pair and add them up:

$$\text{Element} = (a \times x) + (b \times y) + (c \times z)$$

In Fuzzy Max-Min Composition $(R \circ S)$, you replace:

- Multiplication $(\times)$ with the Minimum (min) of each pair.
- Addition $(+)$ with the Maximum (max) of those results.

$$\boxed{\text{Formula: } T_{ij} = \max_k \left(\min \left(R_{ik}, S_{kj} \right) \right)}$$

- R has size 2×3 (2 rows, 3 columns).
- S has size 3×2 (3 rows, 2 columns).
- The resulting matrix T will have size 2×2 (2 rows, 2 columns):

$$T = \boxed{\begin{matrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{matrix}}$$

Step-by-Step Calculation

① Finding T_{11} (Row 1 of R with Column 1 of S)

- Row 1 of R: $[0.2 \quad 0.5 \quad 0.8]$
- Column 1 of S: $\begin{bmatrix} 0.6 \\ 0.5 \\ 0.8 \end{bmatrix}$

1. Pairwise minimums:
   - $\min(0.2, 0.6) = 0.2$
   - $\min(0.5, 0.5) = 0.5$
   - $\min(0.8, 0.8) = 0.8$
2. Take maximum of these minimums:
   - $T_{11} = \max(0.2, 0.5, 0.8) = \boxed{0.8}$

② Finding T_{12} (Row 1 of R with Column 2 of S)

- Row 1 of R: $[0.2 \quad 0.5 \quad 0.8]$
- Column 2 of S: $\begin{bmatrix} 0.9 \\ 0.2 \\ 0.4 \end{bmatrix}$

1. Pairwise minimums:
   - $\min(0.2, 0.9) = 0.2$
   - $\min(0.5, 0.2) = 0.2$
   - $\min(0.8, 0.4) = 0.4$
2. Take maximum of these minimums:
   - $T_{12} = \max(0.2, 0.2, 0.4) = \boxed{0.4}$

③ Finding T_{21} (Row 2 of R with Column 1 of S)

- Row 2 of R: $[0.7 \quad 0.4 \quad 0.3]$
- Column 1 of S: $\begin{bmatrix} 0.6 \\ 0.5 \\ 0.8 \end{bmatrix}$

1. Pairwise minimums:
   - $\min(0.7, 0.6) = 0.6$
   - $\min(0.4, 0.5) = 0.4$
   - $\min(0.3, 0.8) = 0.3$
2. Take maximum of these minimums:
   - $T_{21} = \max(0.6, 0.4, 0.3) = \boxed{0.6}$

④ Finding T_{22} (Row 2 of R with Column 2 of S)

- Row 2 of R: $[0.7 \quad 0.4 \quad 0.3]$
- Column 2 of S: $\begin{bmatrix} 0.9 \\ 0.2 \\ 0.4 \end{bmatrix}$

1. Pairwise minimums:
   - $\min(0.7, 0.9) = 0.7$
   - $\min(0.4, 0.2) = 0.2$
   - $\min(0.3, 0.4) = 0.3$
2. Take maximum of these minimums:
   - $T_{22} = \max(0.7, 0.2, 0.3) = \boxed{0.7}$

Final Answer:

$$T = R \circ S = \boxed{\begin{bmatrix} 0.8 & 0.4 \\ 0.6 & 0.7 \end{bmatrix}}$$